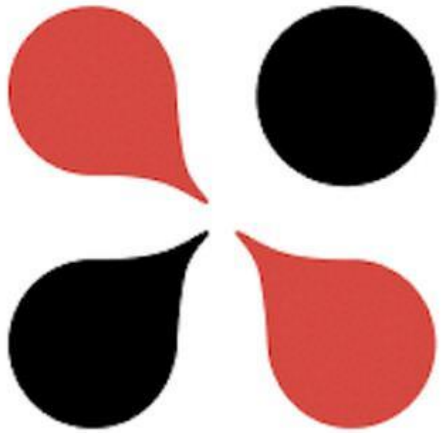
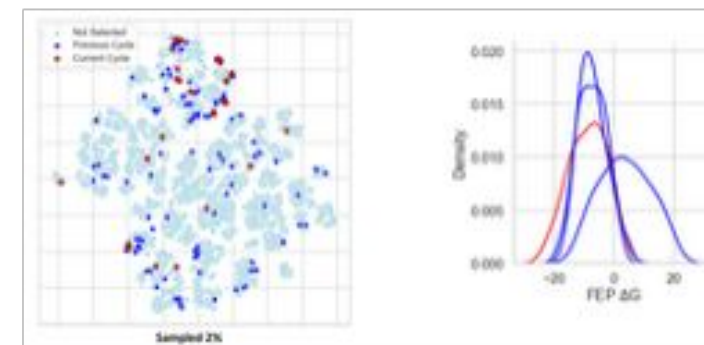
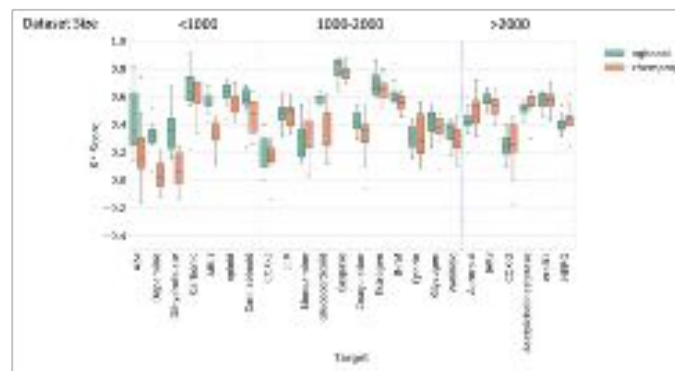
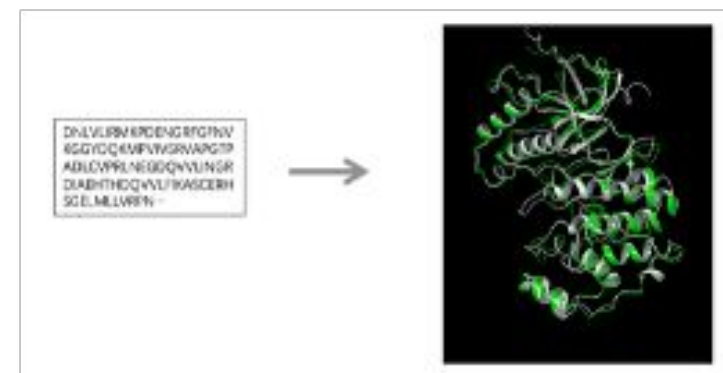
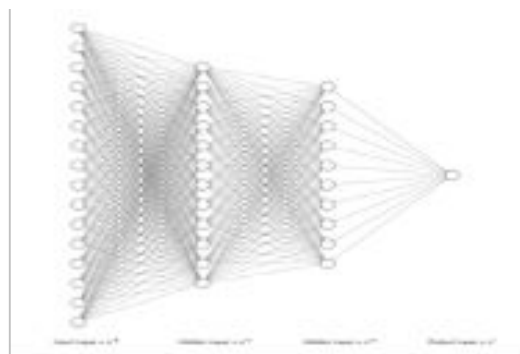




ExpansionRx OpenADMET Blind Challenge

October 27, 2025



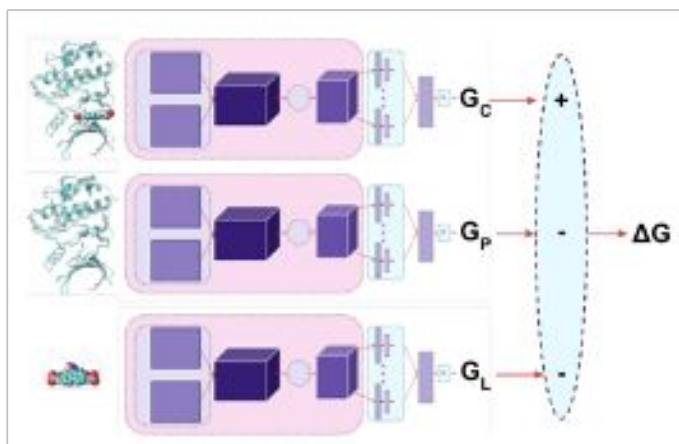
Data is Critical to Machine Learning

Importance

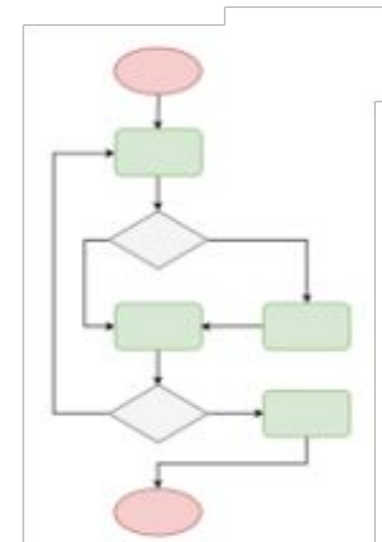
Data

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Representation

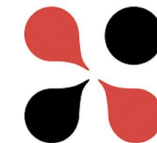


Algorithms



Current Research Efforts

Current Public Datasets Are Highly Flawed

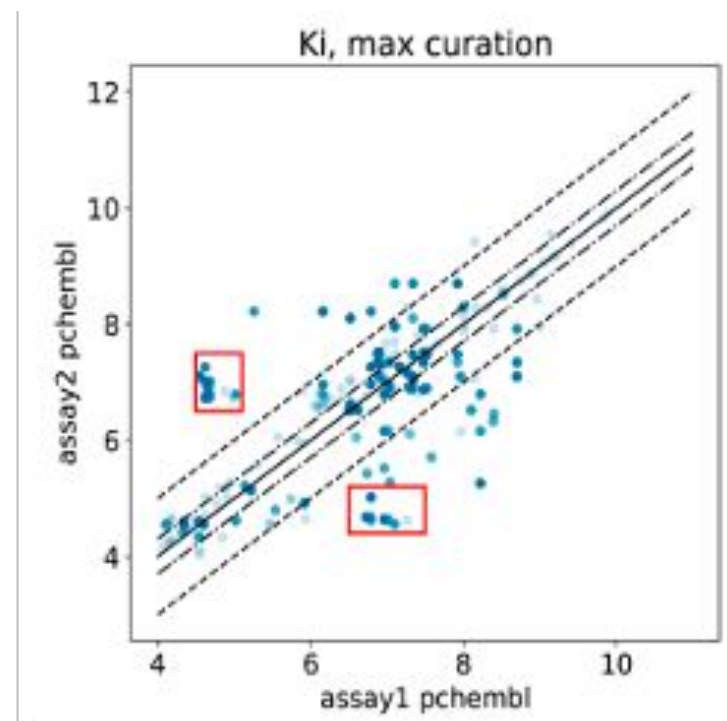
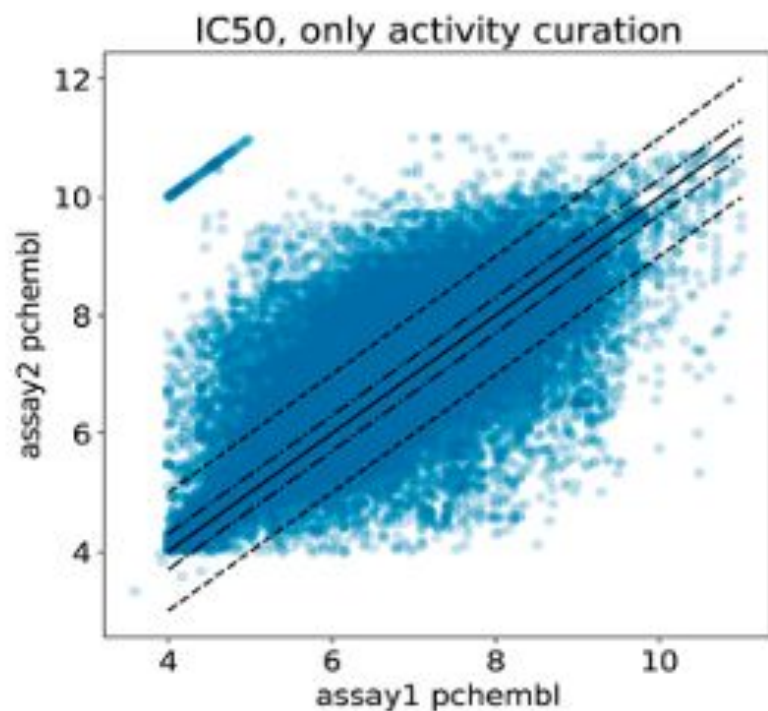


Data collected from dozens of papers, each of which uses different experimental conditions

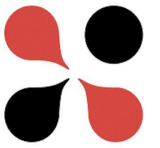
Public data contains numerous curation errors

Dynamic range is often unrealistic

Endpoints are not relevant



Data Generation and Community Engagement Improve ML in Drug Discovery



Improving the understanding of factors that create the most significant delays in drug discovery projects

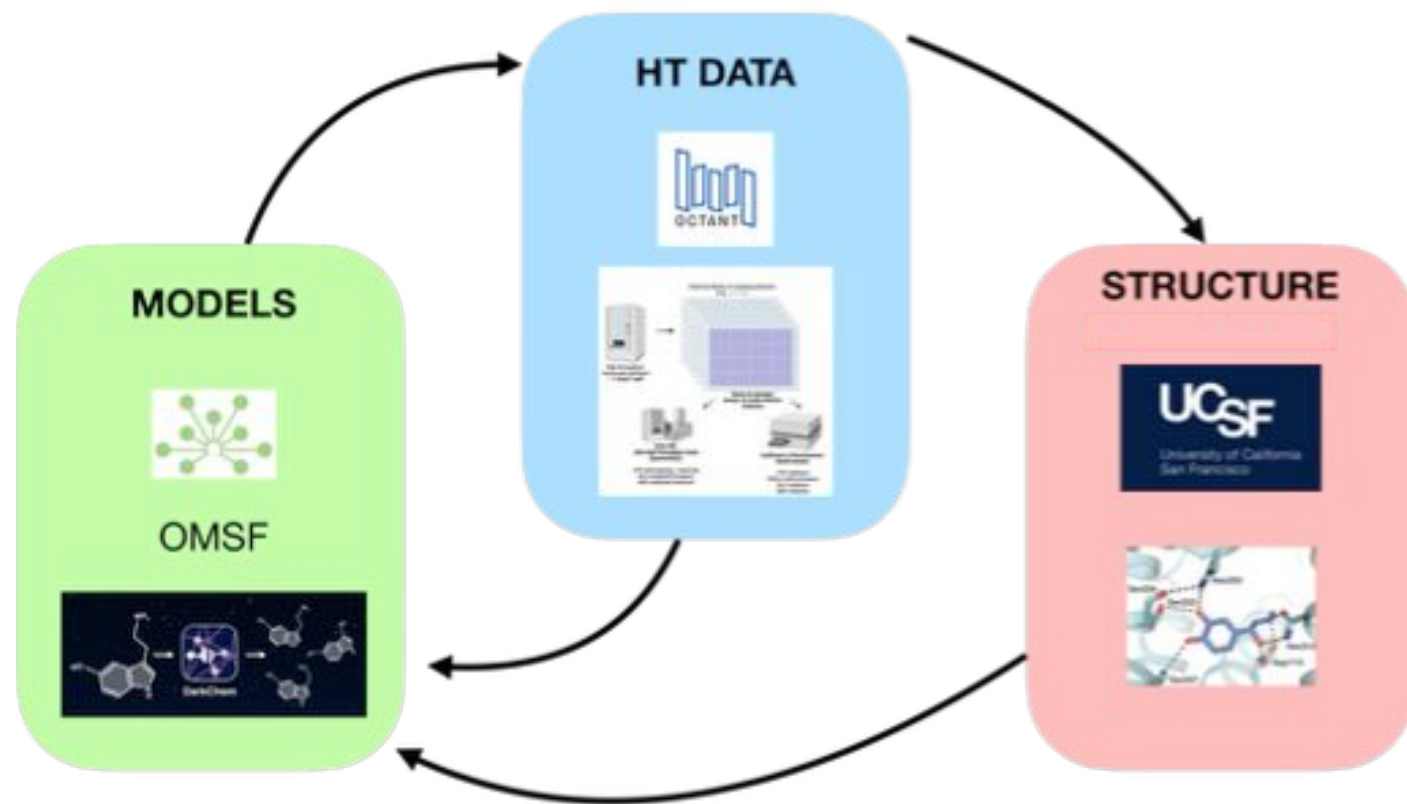
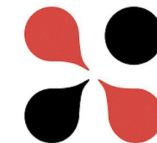
- **Absorption**
- **Metabolism**
- **Excretion**
- **Toxicity**

Generating large high-quality datasets

- **Assay data**
- **X-ray crystallographic and cryo-EM structures of protein-ligand complexes**

Acting as a catalyst for the community

- **Conducting blind challenges**
- **Implementing best practices as open-source software (ANVIL)**
- **Using community feedback to drive data generation and promote best practices**



Gates Foundation



James Fraser
(UCSF)

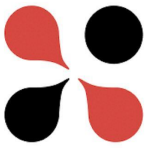


John Chodera
(MSKCC/OMSF)



Sri Kosuri
Octant

ExpansionRx OpenADMET Blind Challenge

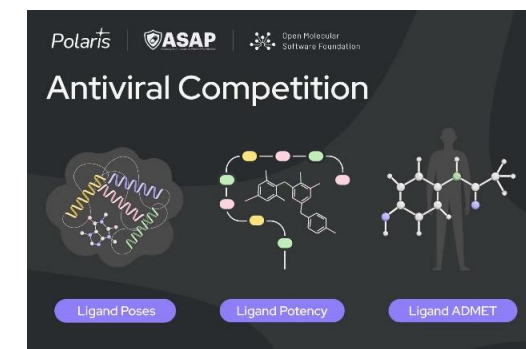
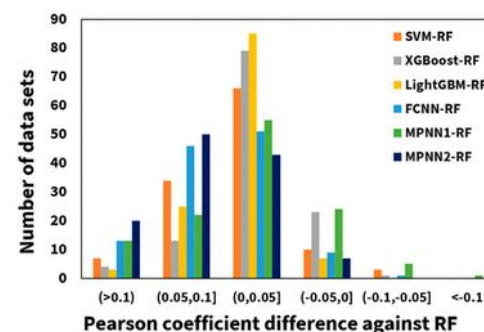
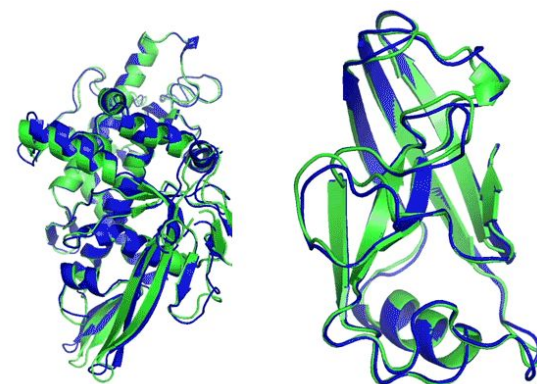
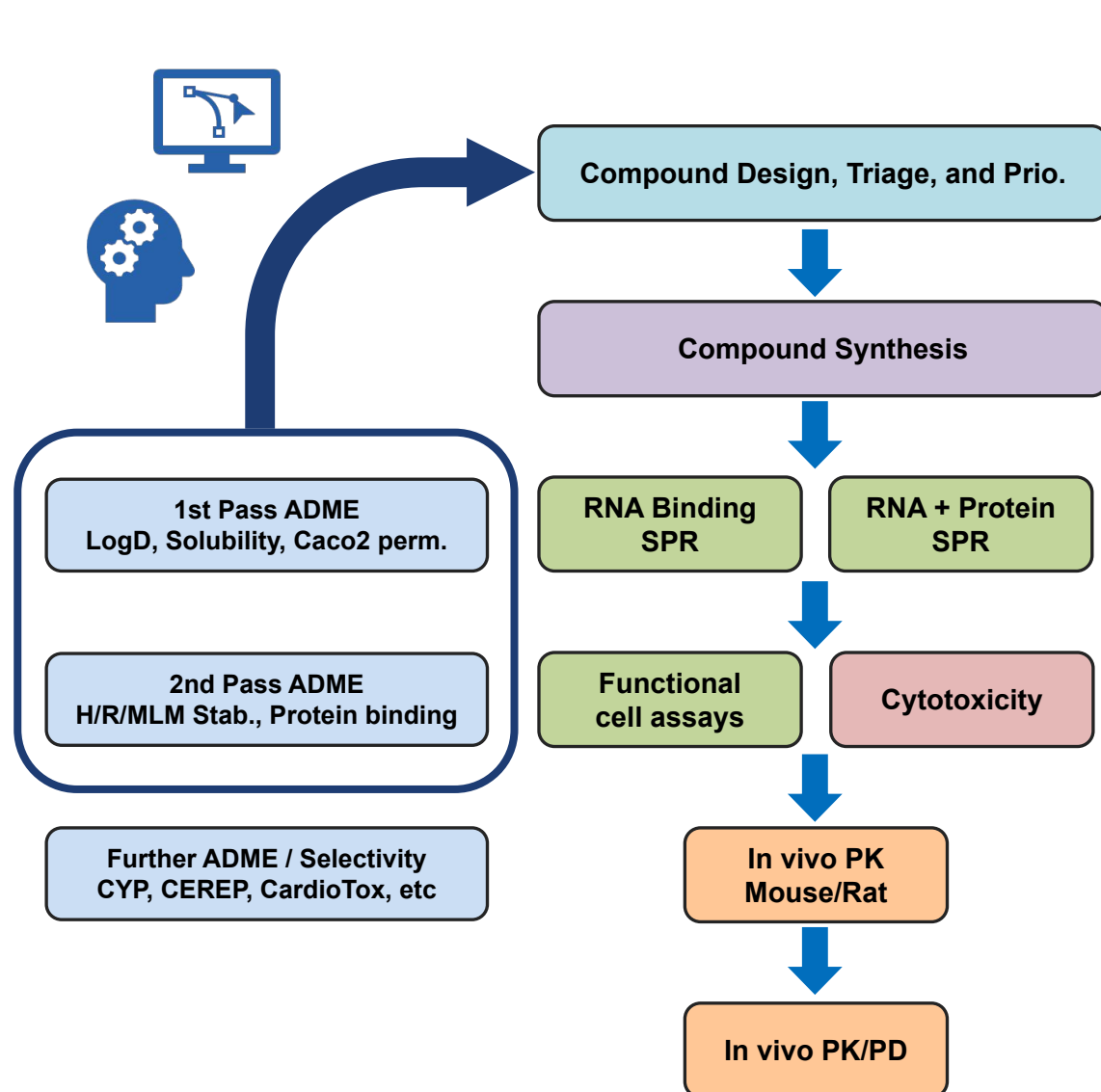
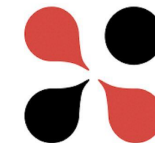


ADMET data from ExpansionRx drug discovery programs

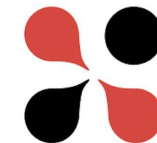
- Realistic time time splits of training and test data

9 distinct assay endpoints

- LogD
- Kinetic solubility (KSOL)
- Human liver microsome (HLM) stability
- Mouse liver microsome (MLM) stability
- Caco-2 Papp A>B
- Caco-2 Efflux Ratio
- Mouse Plasma Protein Binding (MPPB)
- Mouse Brain Protein Binding (MBPB)
- Mouse Gastrocnemius Muscle Binding (MGMB)



Putting ADMET prediction to the test

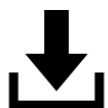


The OpenADMET + ExpansionRx Blind Challenge

- A community-driven effort to benchmark predictive model for drug discovery
- Participants are tasked with developing and submitting their models on a Hugging Face platform



Create a Hugging face account



Download the **public** training dataset

- Hugging Face platform
- CDD platform



Train your model

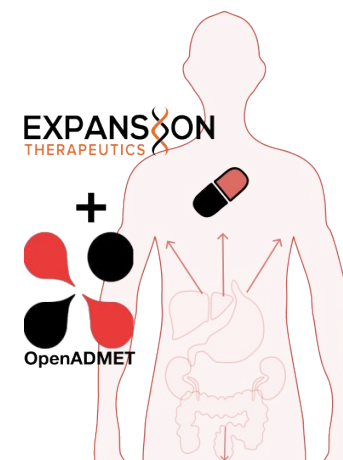


Submit your results & check the leaderboard!

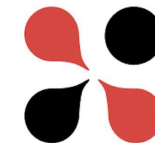


Timeline:

- **October 27: Start of the challenge**
- Oct-Nov: Online Q&A sessions + support via Discord
- **January 19, 2026: Submission closes & final leaderboard**
- January 26, 2026: Winners announced

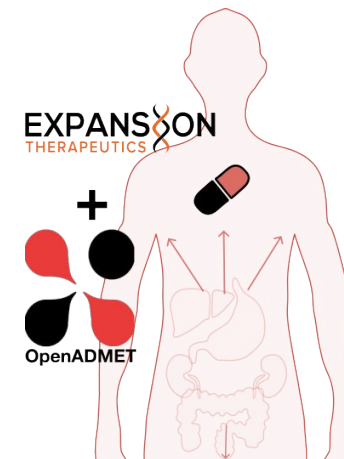


Putting ADMET prediction to the test



More details on the evaluation process

- The leaderboard is **live** and will be updated after every submission.
- You can submit as many times as you want throughout the challenge (**up to a reasonable limit**), with submissions tracked based on Hugging Face username.
- **Live leaderboard $\neq$ final leaderboard:** Evaluation done on different halves of the test set.
- Endpoint ranking is based on MAE, while **global ranking** based on a macro-averaged relative absolute error (**MA-RAE**) across the endpoints.
- **Anonymous participation is possible by providing an *Alias*.**



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Theoretical and Computational Chemistry

A Computational Community Blind Challenge on Pan-Coronavirus Drug Discovery Data

25 July 2025, Version 1

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Working Paper

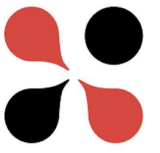
Hugo MacDermott-Opeskin , Jenke Scheen , Cas Wognum , Joshua T Horton ,
Devany West , Alexander Matthew Payne , Maria A Castellanos , Sean Colby,
Edward Griffen , David Cousins , Jessica Stacey , Lauren Reid,
Jasmin Cara Aschenbrenner , Daren Fearon , Blake Balcomb, Peter Marples ,
Charles W F Tomlinson, Duan Li Han, Max Winnick, Haim Rarr, Noa Lahav, Michael Levi



Prizes:

- ➔ All participants are eligible to co-author a future publication detailing the results of the challenge (contact info provided with submission).
- ➔ Participants with notable performance will also be invited to present their work at a special workshop after the conclusion of the challenge.

Ready to join the challenge?



Let's do a tour through the Hugging Face platform 🌟

<https://huggingface.co/spaces/openadmet/OpenADMET-ExpansionRx-Challenge>

The screenshot shows the Hugging Face Space interface for the 'OpenADMET + ExpansionRx-Challenge'. The header includes the 'Spaces' tab, the space name 'openadmet / OpenADMET-ExpansionRx-Challenge', a 'like' button with a count of 6, a 'Running' status indicator, and navigation links for 'App', 'Files', 'xet', 'Community' (with a count of 2), and 'Settings'. The main content area features a welcome message: 'Welcome to the OpenADMET + ExpansionRx Blind Challenge!'. It explains the task: 'Your task is to develop and submit predictive models for key ADMET properties on a blinded test set of real world drug discovery data'. It directs users to the 'Leaderboard' to check progress and to the 'Submit' tab to upload results as a CSV file. A note states: 'The challenge is not yet open for submissions. Please stay tuned for the official launch date!'. On the right, there is a diagram of a human silhouette with a pill icon and the text 'EXPANSION THERAPEUTICS' and 'OpenADMET'. At the bottom, there are tabs for 'About', 'Leaderboard', and 'Submit'.

🌟 OpenADMET + ExpansionRx

Computational Blind Challenge in ADMET

This challenge is a community-driven initiative to benchmark predictive models for ADMET properties in drug discovery, hosted by OpenADMET in collaboration with ExpansionRx.

Why are ADMET properties important in drug discovery?

Step-by-step tutorial also available on Github: <https://github.com/OpenADMET/ExpansionRx-Challenge-Tutorial/tree/main>